UCL DATA ENGINEERING GROUP ASSIGNMENT

Cyber Risk Dashboard

A dashboard built on a full data engineering pipeline using NVD, CISA KEV, EPSS, and cvelistV5.

CVE Rows (Range)	KEV Rate	High EPSS Groups (>=2)
5,133	0.4%	1

Global Controls

Adjust date range and minimum sample threshold for grouped EPSS views.

Start Date: 01/01/2026

End Date: 28/02/2026

Min Scored CVEs (EPSS): 2

Reset Filters

Date range: 2026-01-01 to 2026-02-28 | EPSS groups require scored_cves >= 2

Overview

Key summary metrics generated from the current transformed datasets.

TOTAL CVES
5,133
Rows in the current vulnerability mart

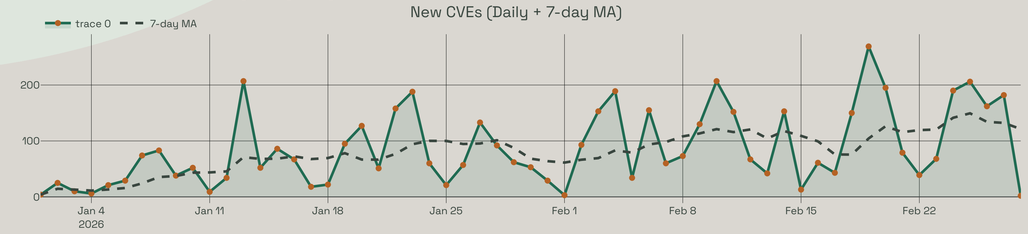
HIGHEST-VOLUME PRODUCT
microsoft / windows_10_1809
171 CVEs in this group

TOP AVERAGE EPSS
ivanti / endpoint_manager_mobile
Average EPSS 0.6247

MOST FREQUENT CWE GROUP
No CWE | samsung / android
795 CVEs in this grouped pattern

Daily New CVEs

Published CVE counts over time with 7-day moving average.



Data Quality Snapshot

Coverage and unknown-rate indicators from transformed analytics tables.

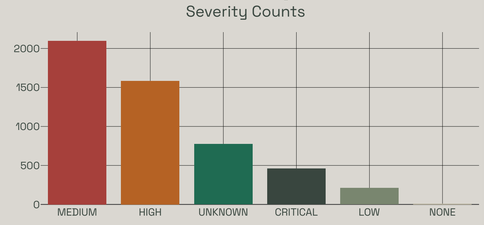
UNKNOWN SEVERITY RATE
15.1%
774 of 5,133 severity rows

UNKNOWN VENDOR/PRODUCT SHARE
0.0%
0 of 3,739 aggregated links

CWE COVERAGE
86.9%
6,451 of 7,421 grouped links mapped to CWE

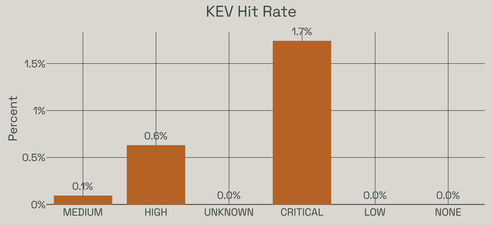
CVSS Severity Distribution

How vulnerabilities are distributed across severity bands.



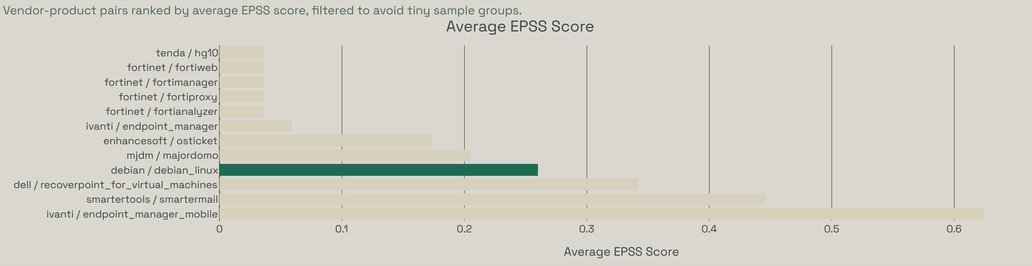
KEV Hit Rate by Severity

Share of CVEs in KEV for each CVSS severity level.



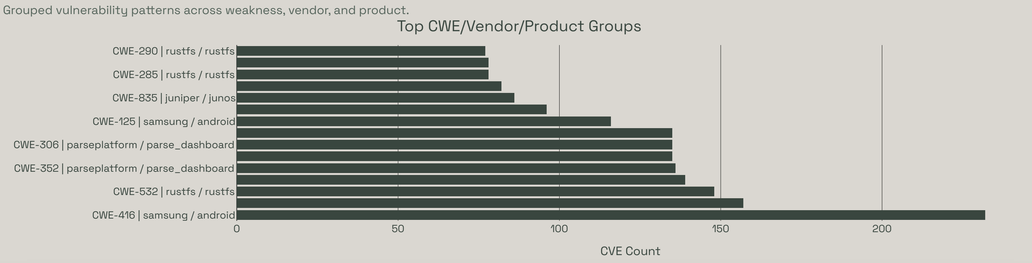
Top Vendor/Product Average EPSS Groups

Vendor-product pairs ranked by average EPSS score, filtered to avoid tiny sample groups.



Top CWE, Vendor, Product Aggregations

Grouped vulnerability patterns across weakness, vendor, and product.



Methodology

How data moves from sources to analytics and visualization.

Raw files are ingested from four public sources (NVD, CISA KEV, EPSS, and cvelistV5) and stored in a raw landing layer. Raw snapshots are then loaded into MongoDB for replayable processing.

Transformation is executed through PySpark and Python ETL to produce curated Parquet datasets, including the vulnerability mart, product bridge tables, star schema dimensions/facts, and aggregation summaries.

Curated Parquet is loaded into DuckDB. The dashboard consumes analytics through FastAPI endpoints (with CSV fallback), and product-level charts are calculated from the CVE-product bridge to preserve many-to-many CVE/product relationships.

Data can be loaded either from static CSV files under `data/curated/transformation_summaries` or from the FastAPI service using `?api_base=http://localhost:8001`.